

Lecture 2 – Data and Statistics

INT104 Artificial Intelligence

Data and Statistics

Dr. Shengchen Li

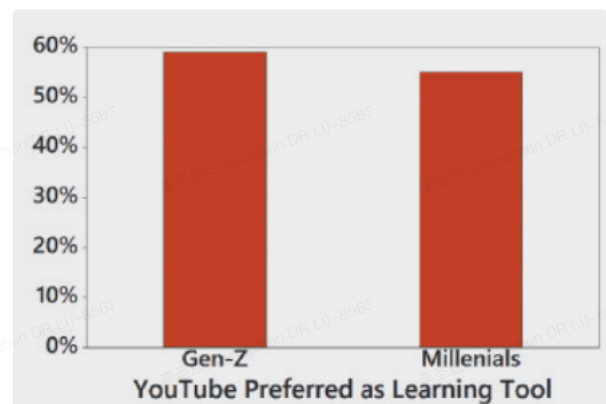
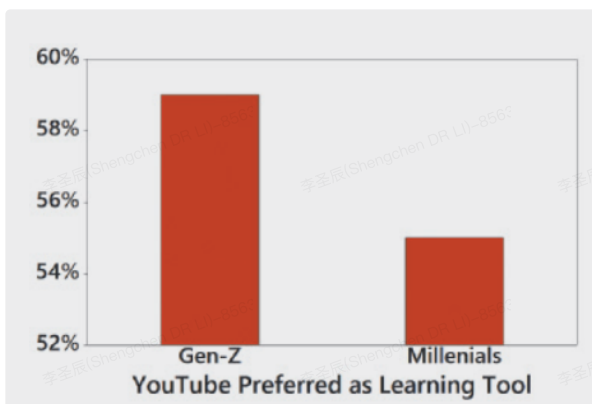
Associate Professor

Department of Intelligence Science

Xi'an Jiaotong–Liverpool University

Critical Thinking and Statistics

- Comparison via Statistics



Now Discuss: Is the difference significant or minor? Why?

Critical Thinking and Statistics

- Sampling and Statistics



In you opinion, how should the data be collected?

- Should we collect the data online?
- Should we collect the data via YouTube?
- Should we collect the data on high street?

Experiment Design Matters



You are going to survey the average mark of INT104. After the exam, you interview your classmates for their score. Is the average mark you obtained reliable?

- Background
 - What's the aim of the study?
 - Is the data collected biased?
 - Is the data representative?
- Analysis
- Conclusion

Terms

- Statistics



What is the relationship between AI and statistics?

- Voluntary response sample



What problems are there with voluntary response sample?

- Survivorship Bias
- Statistical Significance

Potential Threats of Data Collection

- Correlation does not imply causation
- Sample Collection Reported instead of Measured
- Biased Question
- Low Response / Non-sense Response

Data Types



Parameter → Description of a Population
Statistic → Description of a Sample

- Quantitative Vs Categorical
- Discrete Vs Continuous
- Levels of Measurement
- Finite Discrete Vs Infinite Discrete Vs Continuous

Missing Data



What could you do if some data is incomplete?

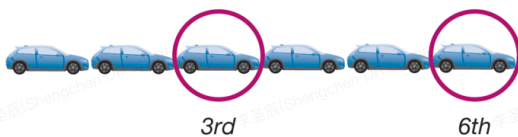
- Delete Cases
- Impute Missing Values



Is that safe and accurate?

Experiment Design

- Replication
- Blindness (and Referenced)
- Randomness
 - Convenience Sampling
 - Systemetic Sampling
 - Stratified Sampling
 - Cluster Sampling



Experiment Design could Consider
Multiples Ways of Grouping



555-867-5309
555-606-0842
555-777-9311



Advanced Experiment Design

- Completely Randomised Experimental Design
- Randomised Block Design
- Matched Pairs Design
- Rigorously Controlled Design

 List a few cases that may cause bias from sampling

Data Observation

Observe the following data

TABLE 2-1 Daily Commute Time (minutes) in Los Angeles																								
18	25	45	75	60	40	25	8	50	10	10	30	15	25	50	20	30	20	45	30	60	30	20	15	30
60	30	15	35	40	5	30	40	20	10	45	30	15	25	25	5	90	30	15	60	20	60	30	25	25

Now discuss how to describe the data:

- Centre
- Variation
- Distribution → Histogram
- Outliers
- Time

TABLE 2-2 Daily Commute Time in Los Angeles

Daily Commute Time in Los Angeles (minutes)	Frequency
0–14	6
15–29	18
30–44	14
45–59	5
60–74	5
75–89	1
90–104	1

Histogram

 How do we decide the boundaries of bins?

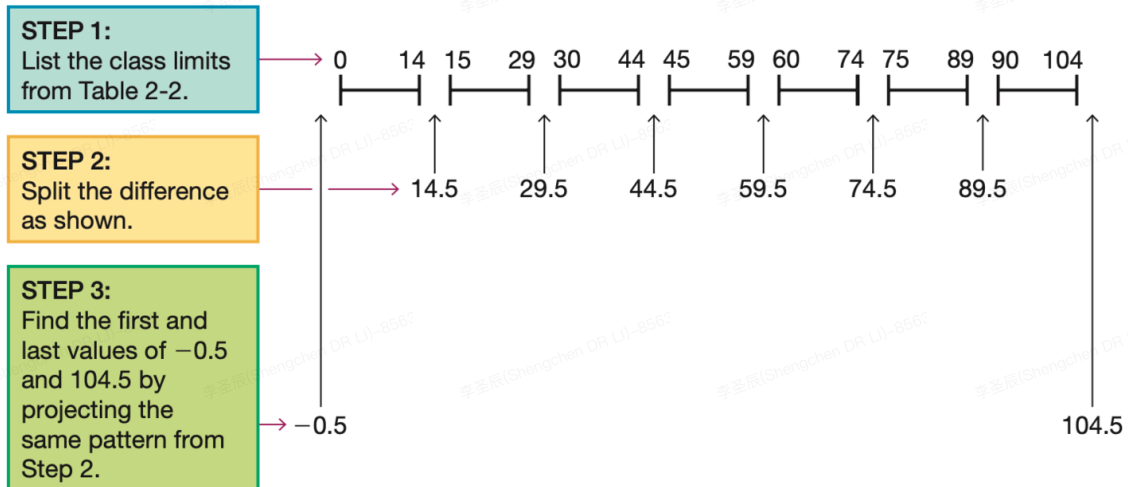


FIGURE 2-1 Finding Class Boundaries from Class Limits in Table 2-2

- How do you use statistics to show the relative length of commute time?

Histogram

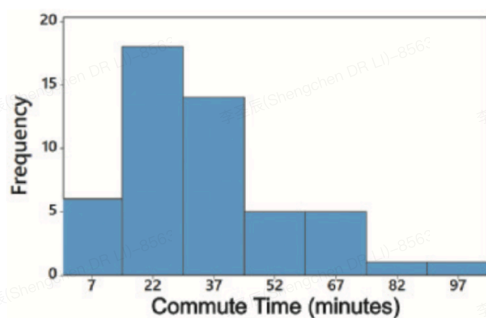


FIGURE 2-2 Histogram of Commute Time in Los Angeles from Table 2-1

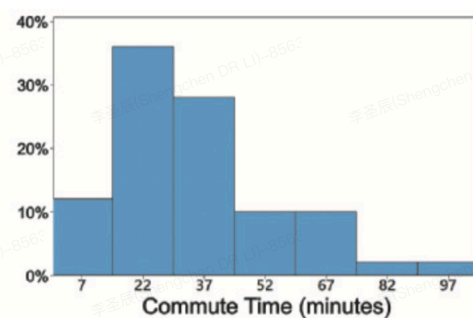


FIGURE 2-3 Relative Frequency Histogram of Commute Time in Los Angeles from Table 2-1

Common Distribution

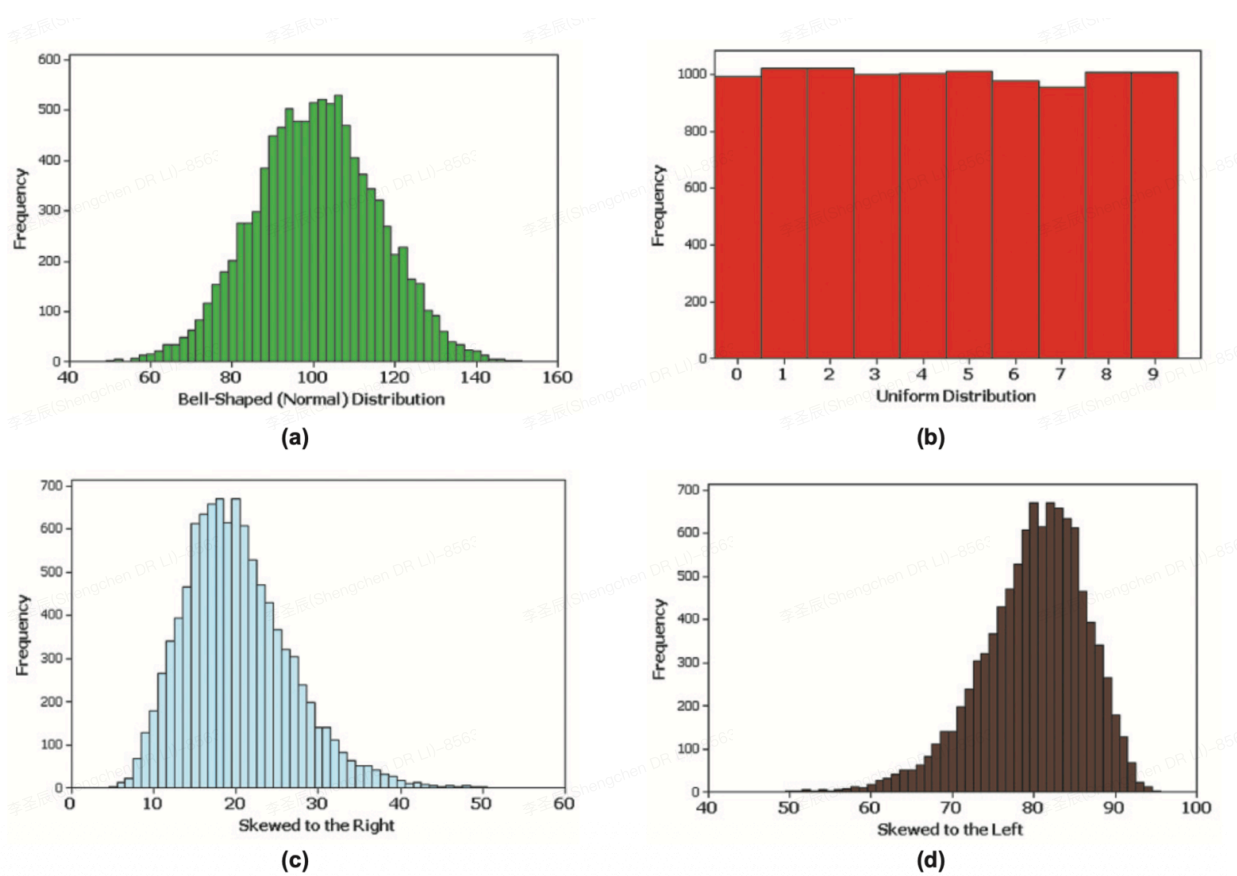


FIGURE 2-4 Common Distributions

Normal Distribution

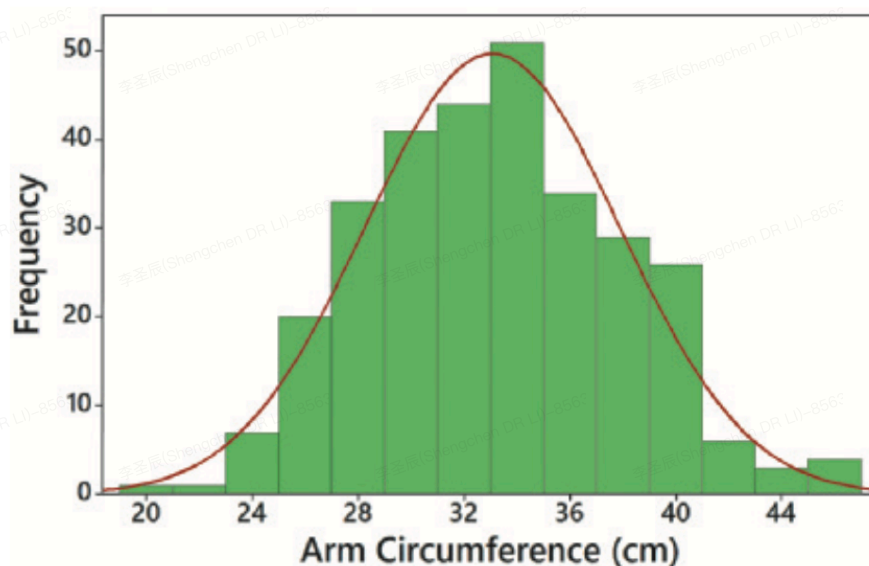
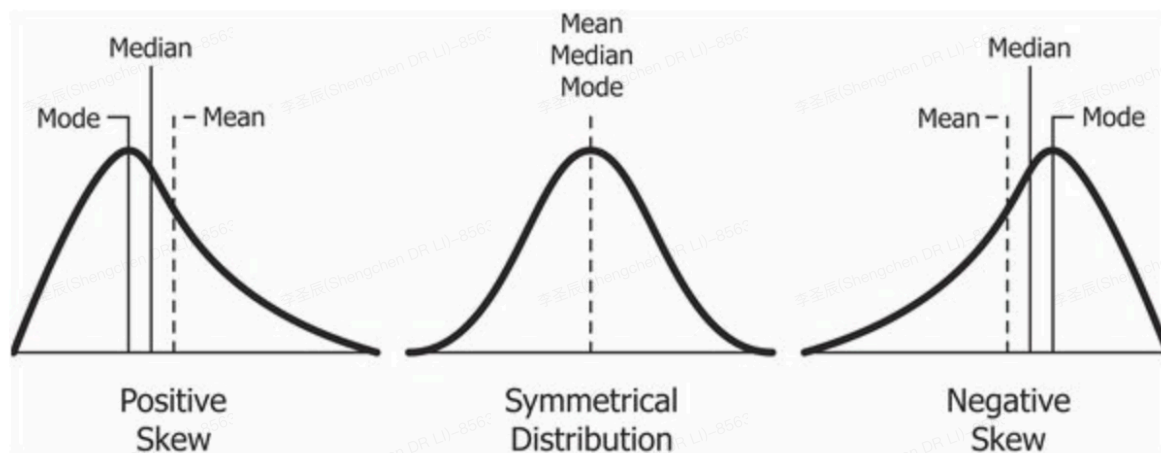


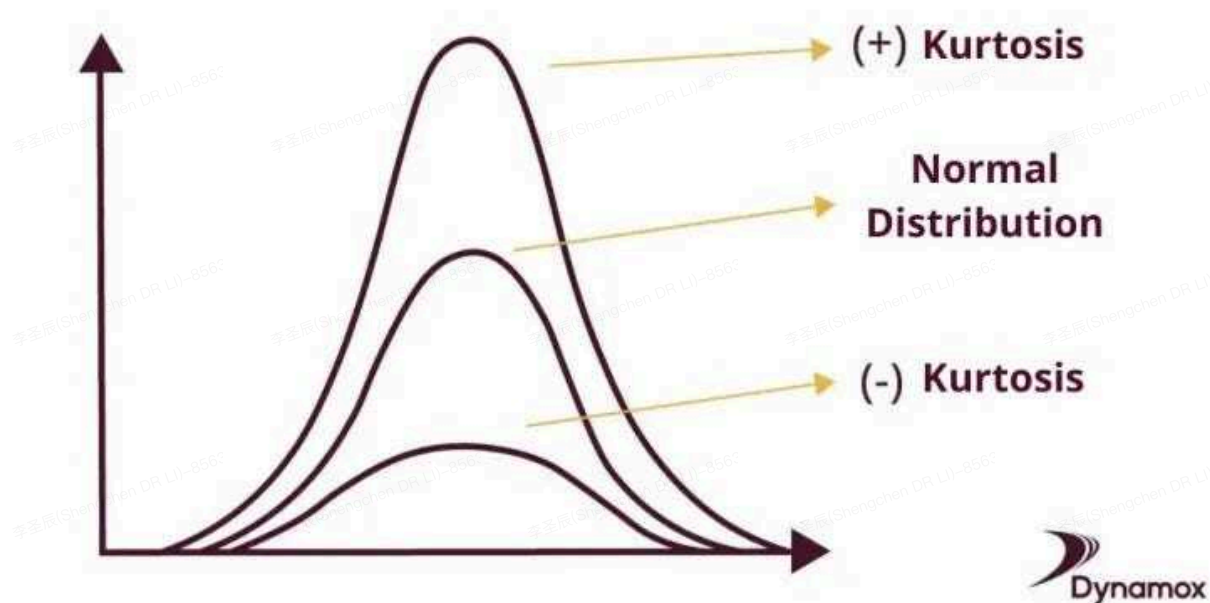
FIGURE 2-5 Bell-Shaped Distribution of Arm Circumferences

Because this histogram is roughly bell-shaped, we say that the data have a *normal distribution*. (A more rigorous definition will be given in Chapter 6.)

Skewness

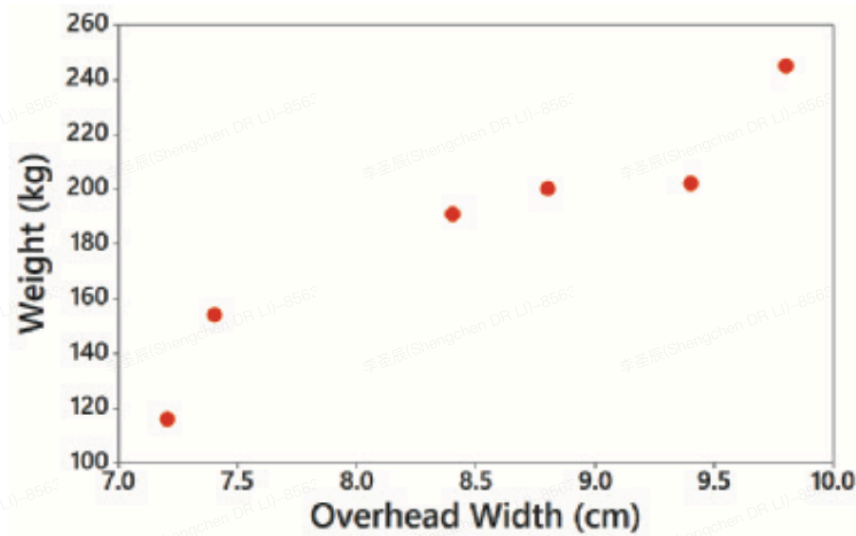


Kurtosis

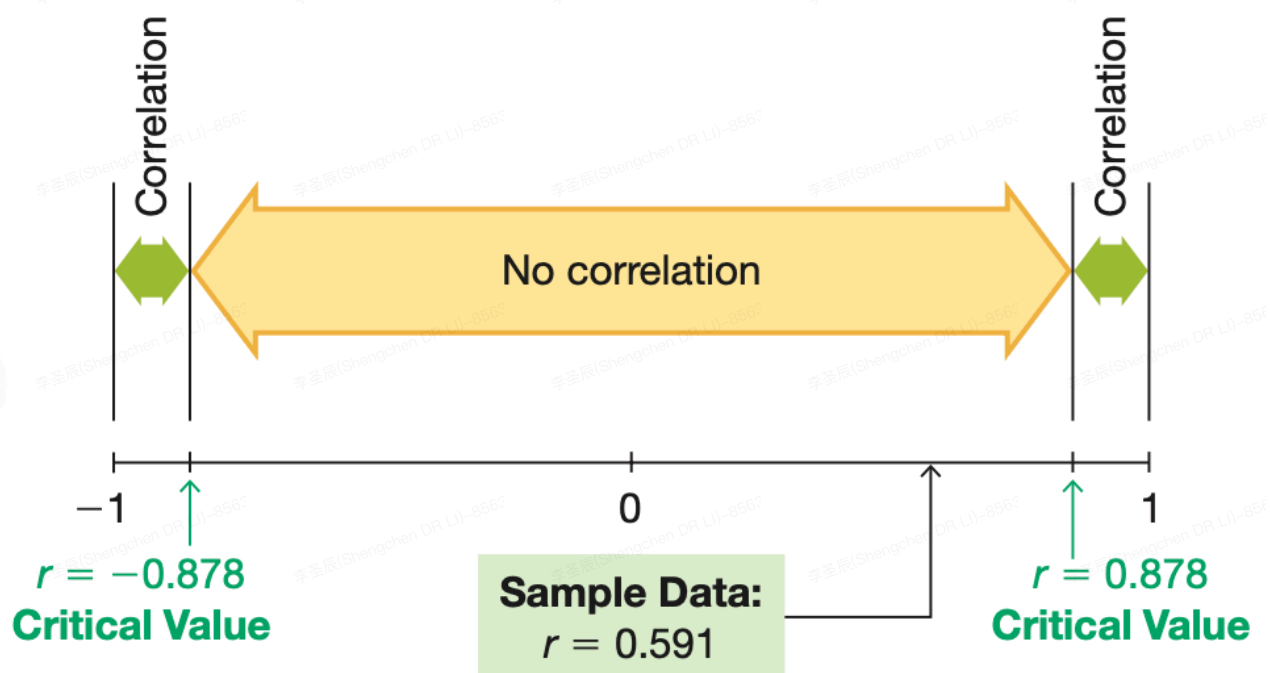


Scatter Plot

Overhead Width	7.2	7.4	9.8	9.4	8.8	8.4
Weight	116	154	245	202	200	191



Correlation Coefficients



Regression

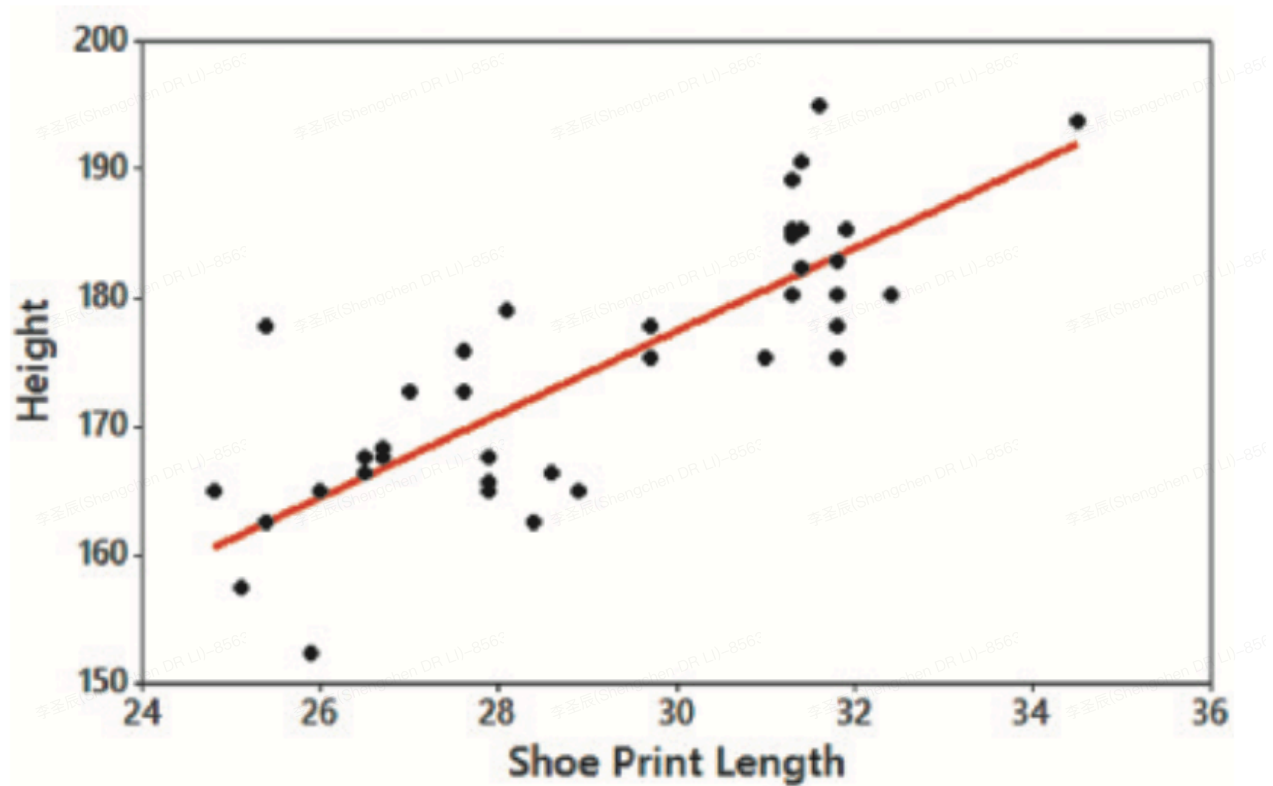


FIGURE 2-19 Regression Line

Boxplot

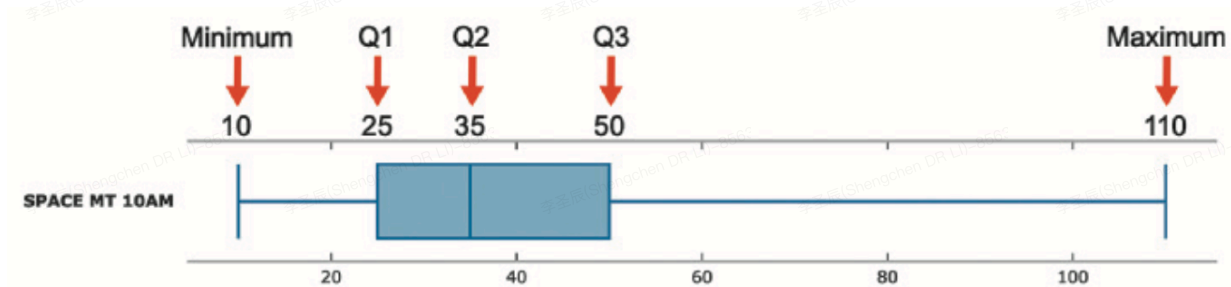


FIGURE 3-8 Boxplot of "Space Mountain" Wait Times (minutes)

Boxplot

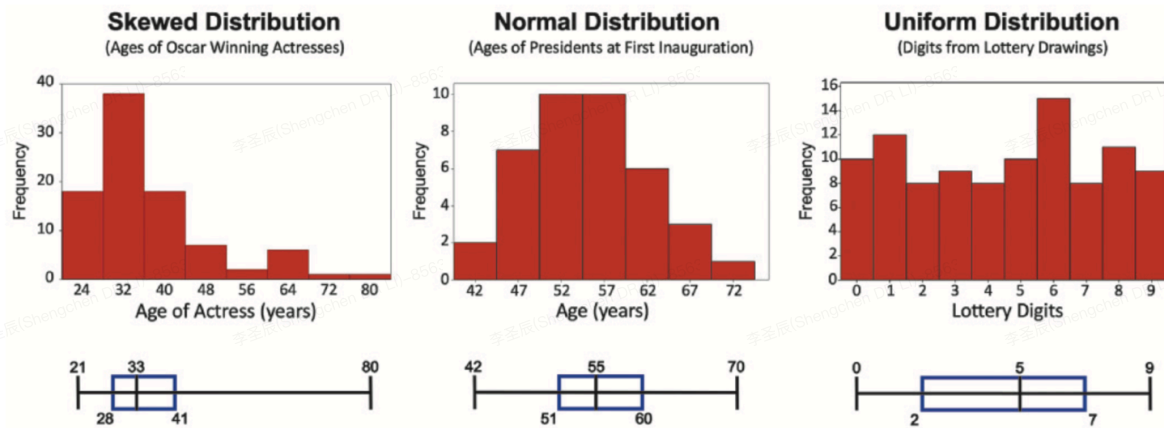


FIGURE 3-10 Histograms and Corresponding Boxplots

Boxplot with Outliers

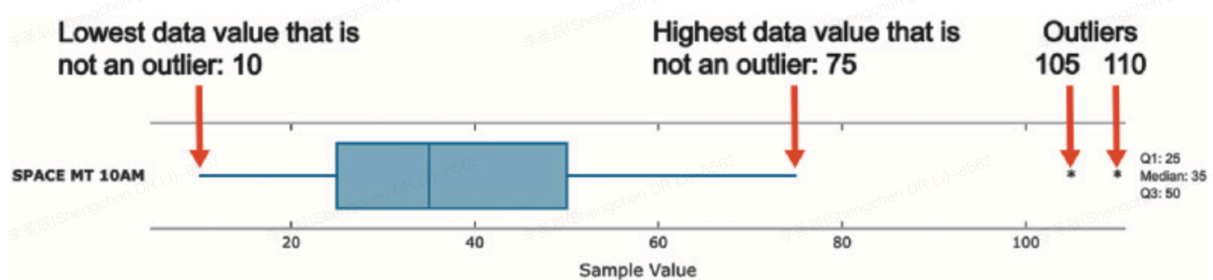


FIGURE 3-11 Modified Boxplot of "Space Mountain" 10 AM Wait Times